

Semester: VII						
COMPUTER GRAPHICS AND VIRTUAL REALITY (Theory and Practice)						
Course Code	:	CS373IA		CIE	:	100+50 Marks
Credits: L:T:P	:	3:0:1		SEE	:	100+50 Marks
Total Hours	:	45L+30P		SEE Duration	:	03 + 03 Hours

Unit-I		9 Hrs
Introduction to Computer Graphics and Virtual Reality: Application areas of Computer Graphics, Introduction to Graphics Programming with OpenGL, The OpenGL API: Graphics Functions, The Graphics Pipeline and state Machines, The OpenGL Interface, Primitives and Attributes, Polygon Basics: polygon types in OpenGL, Attributes, Color, RGB Color, Indexed Color, Control Functions, The Three- Dimensional Sierpinski Gasket. Display Lists Definition and execution of display Lists, Programming. Introduction to Virtual Reality: The three I's of virtual reality, commercial VR technology and the five classic components of a VR system.		
Unit – II		9 Hrs
Input and Output Devices: Input and Interaction: Input Devices. Physical Input Devices, Logical Devices. Measure and trigger. Input Modes. Event-Driven Input: Using the pointing device, Window events, and Keyboard events. Menus. VR related Input Devices: Trackers, Navigation, and Gesture Interfaces; VR related Output Devices: sound displays & haptic feedback		
Unit –III		9 Hrs
Raster graphics algorithms and Geometric Transformations: Points and lines, line drawing algorithms, mid-point circle algorithm; Filled area primitives: Scan line polygon fill algorithm, boundary-fill and flood-fill algorithms. 2-D viewing: The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions, Cohen-Sutherland and Liang Barsky line clipping algorithms, Sutherland –Hodgeman polygon clipping algorithm. Geometric Transformations: 2-D geometrical transformations: Translation, Scaling, Rotation, reflection and shear transformations, matrix representations and homogeneous coordinates, composite transforms, transformations between coordinate systems.		
Unit –IV		9 Hrs
Geometric Transformations (contd..) 3-D Geometrical Transformations: Translation, Scaling, Rotation. Viewing and Visible Surface Detection: Viewing pipeline: viewing coordinates, Aspect Ratio and view ports, view volume, 3-D clipping. Projections: Classification of planar geometric projections, Projections in OpenGL; Visible surface detection: Classification, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.		
Unit –V		9 Hrs
Modelling in Virtual Reality: Geometric modelling: Virtual Object Shape, Object Visual Appearance, Kinematics Modelling: Homogeneous Transformation Matrices, Object Position, Transformation Invariants, Object Hierarchies, Viewing the Three-Dimensional World, Physical Modelling: Collision Detection, Surface Deformation, Force Computation, Force Smoothing and Mapping, Haptic Texturing, Behaviour modelling		

Course Outcomes: After completing the course, the students will be able to	
CO1:	Apply the basic concepts of Computer Graphics & Virtual Reality which illustrates the use of the pipeline architecture, OpenGL library, VR tools.
CO2:	Analyze and make an appropriate choice of methods required for computer representation of 2D/3D objects and modelling in Virtual Reality.
CO3:	Design Computer Graphics and Virtual Reality applications using OpenGL library & VR tools.
CO4:	Implement common geometric construction & VR techniques and develop Computer Graphics/ VR based solution to Engineering applications / real-world problems.
CO5:	Exhibit team work and effective oral/written communication skills to accomplish a common goal of solving Computer Graphics/Virtual Reality problems with the engineering community and society at large,

Reference Books	
1	Computer Graphics with OpenGL, Donald D. Hearn, M. Pauline Baker, Warren Carithers, 4 th Edition, 2011, Pearson Education, ISBN-13 : 978-0136053583
2	Interactive Computer Graphics: A Top-Down Approach Using OpenGL, Edward Angel, 5 th Edition, 2010, Pearson Education, ISBN: 9780321535863
3	Computer Graphics, Zhigang Xiang and Roy Plastock, 2 nd Edition, 2007, ASIN: 0070601658, Tata McGraw-Hill, ISBN-13: 978-0070601659.
4	Burdea, G. C. and P. Coffet. Virtual Reality Technology, 2 nd Edition. Wiley-IEEE Press, 2003/2006, ISBN-13 : 978-0471360896
5	Alan Craig, William Sherman and Jeffrey Will, Developing Virtual Reality Applications: Foundations of Effective Design, Morgan Kaufmann, 2009, ISBN: 978-0123749437

LABORATORY COMPONENT	
PART-A	
	- Write a program to generate a line using Bresenham's line drawing technique. Consider slopes greater than one and slopes less than one. User must able to draw as many lines and specify inputs through keyboard/mouse. - Write a program to generate a circle using Bresenham's circle drawing. User can specify inputs through keyboard/mouse. - Design and develop an OpenGL application program to create two windows. Draw a rectangle of specified width and height by setting four different colors to its corners in the first window. In the second window, draw the same rectangle and spin it continuously. Use the double buffer concept. - Design and develop C program using OpenGL libraries to create two windows: display a cylinder in one window and parallelepiped in second window. Note: Create a cylinder and a parallelepiped by extruding a circle and quadrilateral respectively. - Write a program to recursively subdivides a tetrahedron to form 3D Sierpinski gasket. The number of recursive steps is to be specified at execution time. - Write a program to demonstrate the approximation of a 3D sphere with appropriate mathematical formulations. Write the complete C program to approximate a sphere using OpenGL primitives. - Write a program to fill any given 2D polygon using Scan-line area filling algorithm. - Write a program to implement the Cohen Sutherland line clipping algorithm. Make provision to specify the input for multiple lines, window for clipping and viewport for displaying the clipped image.

9. Write a program to **implement the Liang-Barsky line clipping algorithm**. Make provision to specify the input for multiple lines, window for clipping and viewport for displaying the clipped image.
10. Write a program to **implement the Cohen-Hodgeman polygon clipping algorithm**. Make provision to specify the input polygon and window for clipping.
11. Write a program to **create a house like figure and perform the following operations**.
 - i. Rotate it about a given fixed point using OpenGL transformation functions.
 - ii. Reflect it about an axis $y=mx+c$ using OpenGL transformation functions.
12. Write a program to **create a color cube and spin it** using OpenGL transformations.
The output must be adjusted suitably when the window is resized or moved to a new position.

PART-B

Practice Programs on Virtual Reality (Use VR Tools like Unity 3D, Ureal Engine, Blender):

1. Create virtual objects (cube, sphere, cylinder) and apply materials to simulate realistic appearance.
2. Simulate object movement using translation, rotation, and scaling via transformation matrices.
3. Simulate collision detection between two objects and apply a reaction (e.g., bounce, stop, change color).

Open-Ended learning is to be demonstrated by Case study of any Virtual Reality tools (development of a small application or mini project using VR tools).

Case study of any Virtual Reality tool and Implementation

- Students to explore the tools which provide a VR or 360-degree experience at the primary, elementary and secondary levels.
- Demonstrate the Case-Study implemented.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (40 Marks) and lab test (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE 50 MARKS	50
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		150

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
Student is required to perform Computer Simulation/ Develop a prototype or model as the case may be and present the results in the form of a presentation. Further, students have to submit a poster for exhibition and also a report.		
1	Design and development of the project	20
2	Presentation of the working model/simulation results/prototype building	20
3	Viva Voce	10
TOTAL		50